ISE542 Managing Knowledge

Lecture 7 & 8 – Knowledge Auditing

Complete English Transcript of the Lecture
(Cleaned and structured from audio Part 1 / Part 2 / Part 3)

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Sources: ISE542_Lecture8_Notes.pdf + ISE542_W08_Part1.m4a, Part2.m4a, Part3.m4a. Minor transcription corrections for clarity; content remains faithful to the original lecture.

1. Learning Objectives

- Differentiate between a knowledge audit and an information audit
- The role and importance of a knowledge audit
- The knowledge audit process
- The benefits of conducting a knowledge audit and the hurdles involved
- Various types of knowledge audit methods
- Tools for carrying out a knowledge audit
- KM related planning

2. Two Types of Knowledge Audit: As-is vs To-be

As-is Knowledge Audit: The company wants to understand the current situation – what knowledge already exists, how knowledge processes work, who the knowledge suppliers and knowledge customers are. It is a baseline / stock-taking exercise of the present state.

To-be Knowledge Audit: The audit aims at the future. For example, a company may initiate a knowledge audit to identify the knowledge it needs to develop in the next five years in order to shape its future. It is forward-looking and linked to strategic goals.

In practice, an As-is audit is fundamental. You need a baseline of existing knowledge before you can identify gaps and plan what the company does not yet have but needs for the future.

3. Data Collection Methods

There is no single best method. The auditor must choose a practical combination that balances depth, coverage, time and cost.

3.1 Survey / Questionnaire

Useful for a quick scan of basic knowledge across many people (can be sent to 100 people). However, it only collects limited, structured information; respondents rarely volunteer deep or unexpected knowledge.

3.2 Individual Interview

Best for people whose knowledge is fundamentally different (e.g. department heads of HR, Finance, Production). Their knowledge domains differ, so a focus group would be ineffective. Interviews can surface tacit, in-depth knowledge, but are time-consuming. Senior interviewees (e.g. CEO) are often highly experienced and can bypass poorly structured questions.

3.3 Focus Group

Effective when many people share similar backgrounds (e.g. 90 operators working in similar teams/shifts). Select representative samples (at least one from each team) instead of interviewing everyone. Interviewing 90 people would consume roughly 45 hours of productive time – companies will not allow it, and the auditor would be exhausted. A well-chosen focus group can capture about 80% of the same knowledge with far less time. Focus groups work because people in similar roles tend to hold largely overlapping knowledge.

3.4 Observation

Especially important for skill-based or laboratory work. People may say “I know how to do it”, but when observed they cannot perform. Observation is more reliable than self-report for operational and laboratory skills.

3.5 Practical Combination

For a large organisation: individual interviews for the small management group (different knowledge domains) + focus groups for the large operational population (similar knowledge) + survey for a broad quick scan + observation where skills matter. Always design a practical audit plan that the company can accept in terms of time and disruption.

4. Building the Audit Team – Required Skills

In real organisational life you cannot freely choose teammates – the KM team is already employed by the company. What matters is having the right mix of skills:

- **Strategic understanding** – people who grasp organisational strategy.
- **Active listening skill** – not passive listening (sitting silently and concentrating 100%), but listening while talking, analysing answers in real time and responding. This is significantly harder and not everyone is good at it.
- **Interview technique** – ability to structure unbiased, non-leading questions. Senior people are experienced interviewees and can easily bypass poorly designed questions.
- **Analytical skill** (back-office) – consolidate data from many interviews, spot loopholes, inconsistencies and missing pieces.
- **Coordination** – schedule focus groups and interviews yourself; the client will not prepare the schedule for you.

A failed input (poor data collection) produces a failed output (weak conclusions). Team quality is critical.

5. Knowledge Audit Process Overview

- **Orientation** – explain why the audit is being done and how it will help the organisation (not just capturing knowledge).
- **Culture / KM readiness survey** (quick scan) – “test the water”. If the culture is not ready, more awareness work is needed before forcing KM initiatives. KM is fundamentally a change-management activity; reluctant people will not cooperate.
- Structured data collection using the appropriate mix of methods.
- Produce two core deliverables: **Knowledge Inventory** and **Knowledge Map**.
- Analysis (network analysis, SWOT, Gap analysis, etc.) → Recommendations (short / medium / long term).

Subsequent audits are usually incremental (updating new knowledge) once the baseline has been established. The audit cycle is typically every 6–12 months depending on the company.

6. Knowledge Inventory

A knowledge inventory is a stock-taking exercise – analogous to a physical inventory checklist in logistics. It does **not** record the full detailed content of every knowledge item (that would be enormous and unnecessary). It records the meta-information needed to manage knowledge later:

- Identification / short description of the knowledge item
- Location and format (electronic, paper, etc.) and how to retrieve it
- Owner / supplier (who generates it) and customer / user (who uses it)
- Rating of importance and/or availability. The attributes used for rating must be agreed with senior management; different industries emphasise different attributes (e.g. accuracy for finance, originality for university research, efficiency for government).

Analogy: the contents page of a book – high-level information that helps you decide whether to read the detail. Standardise vocabulary (technique vs method vs tool) so the same item is not listed under different names, which would confuse users.

For tacit knowledge, the communication channel (email, phone, messaging, etc.) is often more important than physical location, because people move around.

7. Knowledge Map and Network Analysis

The knowledge map shows how knowledge flows among stakeholders – who the knowledge centres are, where isolated clusters exist, and the direction of knowledge flows. It is one of the two indispensable outputs of a knowledge audit.

7.1 Knowledge Centres

People (or roles) with intensive knowledge links. They are critical; if they leave, major disruption occurs. KM systems should be designed to support them – reduce their workload of answering repeated requests, enable easy broadcasting of stories and documents, etc. They become key customers of the future KM system.

7.2 Isolated Clusters / Isolated Nodes

Groups or single persons who are weakly connected or completely disconnected from the main network. If they hold critical knowledge (e.g. a machine operator with unique skill, or a specialised lab technician) and leave, that knowledge disappears with them. These must be reported and action taken to connect them or capture their knowledge.

7.3 Unhealthy Knowledge Networks (Multi-step "Levels")

When a knowledge request path involves many unnecessary intermediaries (e.g. You → A → B → C → Keith, when you could ask Keith directly). Such networks make people busy but unproductive. However, do **not** automatically "kill" every multi-step path – always listen to the story behind it before recommending action:

- **Personal dislike / politics** → usually recommend killing the path (it forces other people to be busy for purely personal reasons).
- **Designed gatekeeper / filter** → keep (e.g. an IT manager deliberately uses support staff to filter simple requests so he can focus on strategic work; similar to nurses screening patients in an emergency unit). The design protects the productivity of a critical knowledge holder.
- **Access rights / information security** → usually recommend dissolving the inappropriate path and reviewing whether proper access rights should be granted instead. Compromising information security is not acceptable.

Always report both the network finding and the underlying story. The organisation then decides keep or kill with fuller context.

8. SWOT Analysis and Gap Analysis

8.1 SWOT – Careful Classification

Strength: positive, already existing and verified.
Weakness: already existing and verified deficiency (the company may defend against this claim).
Opportunity: future potential that has not yet been realised.
Threat: a risk that has not yet materialised.

Example: staff turnover rising from 5% to 9% (still single-digit) = Threat. Actual reported turnover of 30% = Weakness (the problem is already happening).

8.2 Gap Analysis – Two Necessary Conditions

A true gap requires **two** conditions simultaneously:
(1) There is a difference between the current state and the desired / required state.
(2) The missing item is **necessary / must-have** for the business.

Missing something that is irrelevant to the business is not a real gap. Example given in class: not knowing C++ is not a gap for this programme because C++ is not required; not knowing IP concepts (patent, copyright, trademark, etc.) is a real gap because it is compulsory for graduation. Companies can successfully defend "we no longer use that knowledge – it is obsolete", so the auditor must check relevance carefully.

Interpretation of explicit vs tacit findings:
- Tacit present, explicit missing → often signals insufficient codification.
- Explicit present, tacit missing → the documents may be obsolete or inactive; they need review or reactivation.
- Both missing for a must-have item → major finding ("jackpot") – the company has a critical knowledge gap.

9. Recommendations and the End of the Auditor's Role

Present each major finding together with a corresponding strategy, clear objective, and suggested duration (short-term ≈ 6 months, medium-term ≈ 1 year, long-term ≈ 3 years – these horizons vary by industry). The auditor's formal role ends at the recommendation stage. Implementation is a separate project; the organisation decides which recommendations to adopt based on its own resources and priorities. It is normal that not all recommendations are taken – that is not a failure of the auditor. Analogy: a physician diagnoses and proposes treatment; the patient decides which treatment to accept.

10. Professionalism in Data Collection

If you record interviews or conversations, you **must** inform the participants in advance and obtain their agreement. Do not hide a recorder in your bag or use a secret camera. That behaviour is unprofessional and can permanently damage trust and future access to the organisation. Place the recording device visibly on the table and explain why recording is needed.

11. Exam-Friendly Key Takeaways

- As-is = current baseline; To-be = future needs. Always start with As-is.
- Method choice depends on knowledge similarity, number of people, time and cost: survey (breadth), interview (depth / different domains), focus group (similar groups), observation (skills).

- Two core outputs of every knowledge audit: Knowledge Inventory (what / where / who / rating) + Knowledge Map (flows, centres, isolates, unhealthy paths).
- Isolated nodes and knowledge centres are high-priority findings that must be reported.
- Unhealthy multi-step paths: investigate the underlying story before recommending keep or kill (politics vs designed gatekeeper vs security).
- A real Gap = difference + must-have for the business.
- SWOT: Weakness = already true; Threat = not yet true.
- The auditor recommends; the organisation decides whether and how to implement.
- Always obtain consent before recording; professionalism is non-negotiable.

— End of Complete English Transcript —

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